Designing Scalable Batch Processing Pipelines for High-Volume Data in Urban Mobility and MOOC Learning Analytics

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1. Introduction

Organizations need to be able to efficiently gather, process and analyse the data in order to be able to generate useful information which can help them make smarter decisions, improve their efficiency as well as user experiences [1]. The data intensive systems are the key in type of large-scale data processing and data analysis activities due to their exploitation of large-scale architectures and sophisticated technologies.

This report design and demonstration of two distinct data ingestion and transformation pipelines:

- Optimization of Urban Mobility using Analysis of Traffic Sensors Data
- Behavioural Analysis in MOOC (Massive Open Online Courses)

The two use cases demonstrate how contemporary data pipelines in addressing the challenges unique to the domain by allowing the collection, transformation, storage, and visualization of the data.

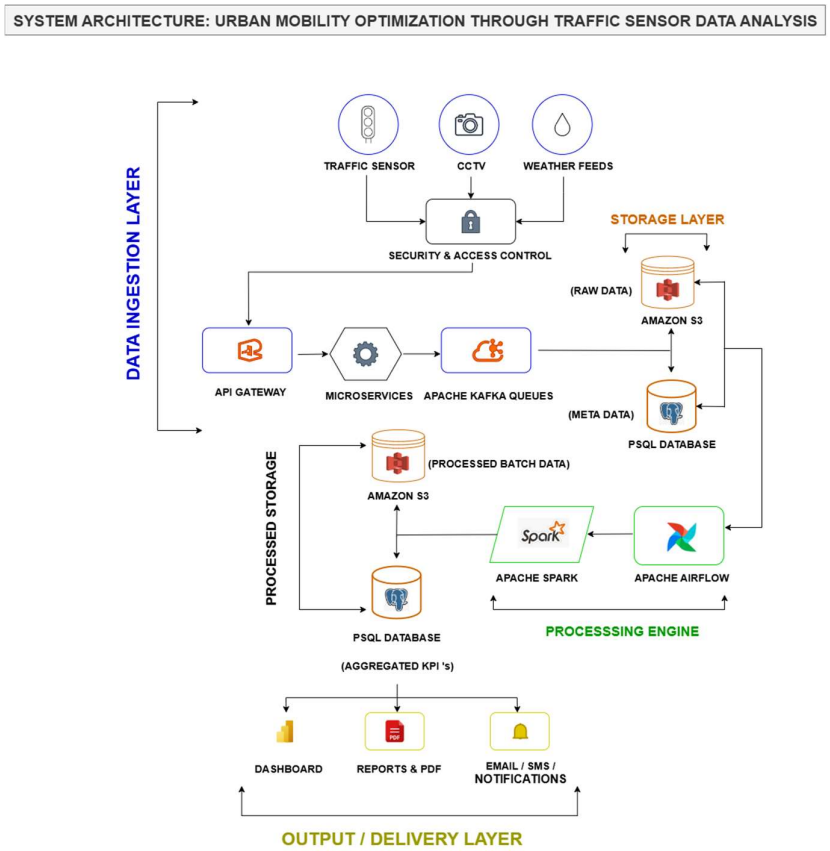
2. Use Case 1: Urban Mobility Optimization through Traffic Sensor Data Analysis:

2.1 Problem Statement

Urbanization expansion and a high number of cars, cities across the globe are suffering by having more traffic jams, pollution and risk of accidents,[2],[3]. The traditional traffic management systems tend to be reactive and unable to process large volumes, as varieties of real-time data sourced by real-time sources like to the traffic signals, CCTV cameras, environmental sensors especially the huge amount of data cannot achieve the comprehensive support and real-time interaction and is not useful in contingency management. The system must be open to process high frequency sensor data, weather updates and incident reports that appears as batches and it should be fault tolerant and high scale which helps the government to make timely data-driven decisions to improve the traffic congestion, decrease environmental pollution and prioritize the safety of the commuters.

2.2 Architecture Diagram

The diagram shows a simplified end-to-end process flow starting and ending at traffic sensors, CCTV feeds and weather APIs entering a secured API Gateway, where stream normalisation via specialised micro-services is performed before passing it to the Apache Kafka to store buffered data at high IO rates. Raw payloads are stored in Amazon S3, reference data persists to PostgreSQL, providing the processing tier quick access to file as well as relational stores. Airflow runs Apache Spark, which does batch aggregations, validation, and outliers checks, storing cleansed metrics in S3 and PostgreSQL. Lastly, dashboards, scheduled PDFs and real-time alerts bring to attention of operations staff these KPIs, bridging the gap between what happens in the street and decisional authority.



(Figure 1: Pipeline Architecture for Urban Mobility Optimization through Traffic Sensor Data Analysis)

Open with draw.io: https://drive.google.com/file/d/1X-O9QdzkCGO3QVCQ21yqj9jYLP-IbTpB/view?usp=sharing

2.3 Technology Justification (Urban Mobility Optimization)

- API Gateway & Microservices: Provides secure and modular ingestion of traffic data. The ability to scale and make faster updates is important to urban evolution and it can be achieved using microservices.
- Apache Kafka [7]: Its technology is simple and its performance demand to stream high-velocity data at scale but with durability and low-latency resulted in the adoption of Apache Kafka that can out-perform message queues effectively.
- Amazon S3[6]: Delivers low cost and unlimited storage for unstructured and semi structured traffic data which can be used in both Remote analysis and Batch processing.
- PostgreSQL: It enables storing of structured data and allows fast querying of incident logs, which adds relational functionalities.
- Apache Spark: It provides high speed performance, process and distribute the data and enables anomaly detection such as congestion spikes or accident patterns.
- Apache Airflow: An orchestra of workflow automation, which carries out complicated dependencies in the scheduled time within a batch pipeline effectively.
- Power BI & Notification System: Power BI is an effective tool for the visualization of data using an interactive dashboard and reports. whereas the alert system initiates the actionable alerts via messaging to the stakeholders.

2.4 Workflow Explanation for (Urban Mobility Optimization)

Data Ingestion: The sensors (JSON), CCTV(CSV), and weather feeds API(JSON) provide real-time traffic data that is ingested in the system through the API Gateway. This data is then applied to specifically dedicated microservices responsible to handle specific data feeds.

Data Storage Amazon S3 stores the raw data, but metadata and events are stored in PostgreSQL in order to incorporate structure lookups and analytics.

Processing: Data streams are stored in Apache Kafka and sent to Apache Spark[7] that cleans, aggregates and identifies patterns (i.e. traffic jam, rush hour). Examples of batch workflows that Apache Airflow takes care of are periodic congestion summaries or predictive modelling, Then the processed data are stored again in Amazon S3 and PostgreSQL respectively.

Output & Delivery:

- Power BI Dashboards - for real time visualization of data
- PDF Reports - for periodic reporting
- Email/SMS Alerts - for critical incident notifications

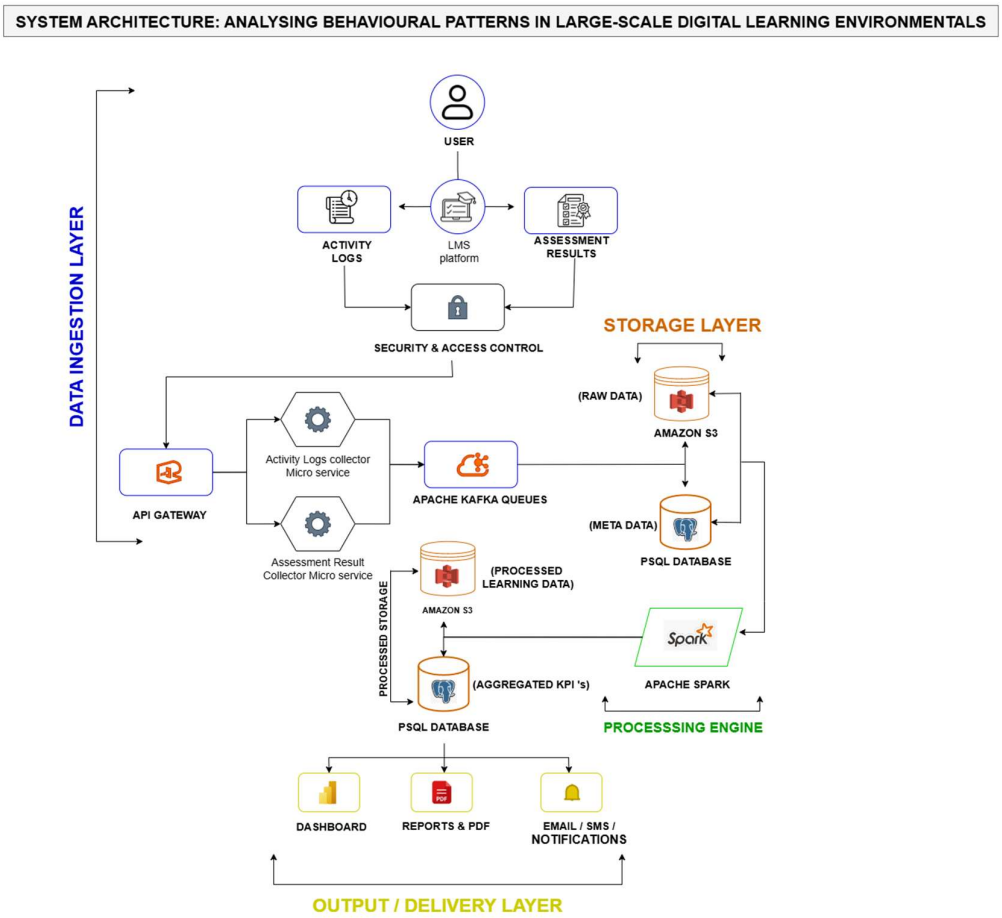
3. Use Case 2: Analysing Behavioural Patterns in Large-Scale Digital Learning Environments (MOOCs)

3.1 Problem Statement

Mass Open Online Courses (MOOCs) have changed the face of the education process since this process enables individuals to learn in a convenient manner[4],[5]. The serious cause of concern to these platforms is the fact that they have to make sense of learner engagement, predict dropouts and make courses more efficient and the wide range of data is extracted from student interaction every day. The problem is that the traditional systems cannot manage and react to huge data on behaviour in an effective manner in real-time. Typical data sources include activity log, reviews, clickstream, and the consumption of video data in bulk and real time. A batch processing system can facilitate the process by grouping and analysing massive amounts of data on learner interactions. Through this MOOCs can be able to make discoveries and forecast dropouts, enhance course designs in an attempt not to overload real-time systems.

3.2 Architecture Diagram

Records of learner activity and assessment outcomes are routed by API Gateway to two small micro services that separate the behavioural and grading data. Apache Kafka pipes the events through Amazon S3, which does buffering of the raw content and PostgreSQL stores structured metadata. Cadence orchestrated Spark jobs process the heterogeneous inputs into engagement KPIs and dropout risk scores and keep the outputs available to query at both stores, interactively. Power BI dashboards auto generated PDF briefs and rule based email/SMS notifications convert these insights to timely interventions on the side of instructors so that the platform can respond accordingly to changing student behaviour rapidly.



(Figure 2: Pipeline Architecture for Analysing Behavioural Patterns in MOOCs)

3.3 Technology Justification (Behavioural Analysis in MOOC)

- API Gateway: Allows in scalable modular ingestion of real time logs and assessment data.
- Microservices: Activity Log Collection and Assessment Result Collection are split through the use of microservices so they can be scaled separately, the development cycle is multifold quicker, and it is easier to isolate faults.
- Apache Kafka [7]: It was selected due to its capacity to manage real-time streaming data with a high degree of reliability and at a large scale, at least compared to the alternatives such as RabbitMQ working better in high throughput environments.
- Amazon S3 [6]: Serves as the main storage of raw, semi-structured and batch data, and is cheaper, elastic, and more durable than the traditional on-premises storage.
- PostgreSQL Database: It is a good choice to store and query structured meta-data and it is ACID compliant.
- Apache Spark: We have chosen it due to its in-memory distributed processing features, which permit it to deal effectively with both batch and stream data, thus delivering better performance compared to the archaic MapReduce models.
- Power BI & Email Notifications: Power BI provides customers with an interactive and low latency dashboard whereas email and SMS makes use of proactive communication to the end-users in order to make sure that the insights are actionable.

3.4 Workflow Explanation (Behavioural Analysis in MOOC)

- Data Ingestion: The users use the LMS platform that generates the logs and assessments. Those are received through API Gateway and forwarded to individual Activity Log Collector and Assessment Result Collector microservices and from there the data is transferred to the Apache Kafka queues to initially buffer information and then to stream it.
- Data Storage: Raw data is stored in Amazon S3, assessment-related metadata is stored in PostgreSQL.
- Data Processing: Apache spark processes the raw and meta-data where it performs aggregations (eg: student engagement KPIs) and the processed data will be stored again in Amazon S3 and PostgreSQL.
- Output & Delivery: Aggregated KPIs are displayed in the dashboards, PDF reports, or trigger notifications.

4. Conclusion

Both data ingestion and transformation pipelines display and demonstrate how scalable and fault tolerant architectures can solve domain specific challenges in urban mobility and digital learning. Through batch processing, organizations can extract actionable insights from large data sources [1]. These pipelines ultimately enhance decision making, optimize operational efficiency and improve user experience.

5. References

[1] T. Parmar, "Scaling data infrastructure for high-volume manufacturing: Challenges and solutions in big-data engineering," Int. Scientific Journal of Engineering and Management, vol. 3, no. 3, Mar. 2024. [Online]. Available:https://isjem.com/download/scaling-data-infrastructure-for-high-volume-manufacturing-challenges-and-solutions-in-big-data-engineering/.

[2] J. Wilbur, M. Pugliese, A. Laszka and A. Dubey, "Efficient data management for intelligent urban-mobility systems," arXiv preprint arXiv:2101.09261, Jan. 2021. [Online]. Available: https://arxiv.org/abs/2101.09261.

[3] D. Teledjieu and I. Shafique, "CityPulse: Real-time traffic data analytics and congestion prediction," arXiv preprint arXiv:2506.01971, May 2025. [Online]. Available: https://arxiv.org/abs/2506.01971.

[4] D. Otoo-Arthur and T. L. van Zyl, "A scalable heterogeneous big-data framework for e-learning systems," in Proc. IEEE Int. Conf. Artificial Intelligence, Big Data, Computing and Data Communication Systems (icABCD), Durban, South Africa, 2020, pp. 1–6. [Online]. Available Accessed via National College of Ireland IEEE Xplore.

[5] K. Palanivel and T. Chithralekha, "Big-data reference architecture for e-learning analytical systems," Int. J. Recent Innov. Trends Comput. Commun., vol. 6, no. 1, pp. 55–67, Jan. 2018. [Online]. Available: https://www.researchgate.net/publication/322626168_Big_Data_Reference_Architecture_for_e-Learning_Analytical_Systems.

[6] Amazon Simple Storage Service (S3) " Developer Guide, Amazon Web Services,"2024. [Online]. Available: https://docs.aws.amazon.com/s3/.

[7] Apache Kafka 3.7 "Documentation, The Apache Software Foundation, " 2025. [Online]. Available: https://kafka.apache.org/documentation/.